

Vibe-IC — ALL Phases / Stages / Steps (v2.2.0, exhaustive)

Plugin 0.2.2. Complete enumeration of every phase, stage, and step. There are **two coexisting views**: **(A)** the runner ground-truth markers (`[N/15]` / `def step_*`) that the code actually prints, and **(B)** the 33-step Stage model (sign-off view). Both are listed in full below; the companion `CANONICAL_FLOW_v2.2.0.md` is the narrative. Source of truth = the runners.

0. Entry + orchestrator + pre-flight

ID	Item	Where
P0	Pre-flight (env / PDK / tool availability)	<code>mcp_server_health_check</code> , <code>eda_doctor</code>
—	Path A: NL prompt / dialogue → Phase 1	<code>phase1_doc_one_shot_runner.py</code>
—	Path B: vendor docs → Phase 1 (docs mode)	same
—	Orchestrator: Phase 1 → Phase 2(=2a+2b) → Analog (after P2) → Phase 3	<code>vibe_ic_one_shot_runner.py</code>

A. RUNNER-MARKER VIEW (ground truth)

ONE continuous global step number across the sequential flow Phase 1 → 2 → 3 (Phase 3 does **NOT** restart at 1 — it is global 50-55). Analog A1-A8 / Mixed M1-M4 run **parallel** to Phase 2 and keep their native A/M ids. Auto-generated live in `FLOW_STEPS_GENERATED.md` (regenerate with `flow_doc_emit.py`). § B below is the orthogonal 33-step silicon-flow model (its own 1→33).

A.1 Phase 1 — global steps 1-34 (15 main markers + 19 sub-steps, `phase1_doc_one_shot_runner.py`)

Marker	Step
<code>[1/15]</code>	Text extraction (input/docs → input_doc; 2 MB scan cap, v0.1.91)
<code>[2/15]</code>	L1_DATASHEET
<code>[3/15]</code>	L2_FRS
<code>[4/15]</code>	L3_CMD_PROTOCOL
<code>[5/15]</code>	L4_REGMAP
<code>[6/15]</code>	L5_ADI_SPEC
<code>[7/15]</code>	L6_CONTROL_LOGIC
<code>[8/15]</code>	L7_TEST_DEBUG
<code>[9/15]</code>	L8_RTL_CONSTANTS
<code>[10/15]</code>	L9_INTEGRATION_SPEC
<code>[11/15]</code>	L10_TEST_CASES
<code>[12/15]</code>	L11_OTP_CONTENT
<code>[13/15]</code>	L12_BEHAVIORAL_SEQUENCES
<code>[14/15]</code>	L13_LAB_CALIBRATION
<code>[14b/15]</code>	L8_TIMING_WAVEFORM
<code>[14b2/15]</code>	L8 protocol-width extract (R19)
<code>[14b3/15]</code>	L8 encoding-table overlay (R41)
<code>[14b4/15]</code>	L6 FSM / control-logic overlay (R42)
<code>[14b5/15]</code>	L12 behavioral-sequences overlay (R43)
<code>[14b6/15]</code>	L17/L18/L8_TIMING/L9 batch synth (R46) — DEPRECATED → 14c3

Marker	Step
[14b7/15]	L8_RTL_CONSTANTS universal protocol constants (R48)
[14c/15]	L14-L18 protocol spec extract
[14c0/15]	L9 integration_spec overlay (R40)
[14c1/15]	L1 protocol metadata overlay (R23)
[14c1b/15]	L17 handshake_pairs overlay (R27)
[14c2/15]	L3 protocol mirror from L14-L18 (R21)
[14c3/15]	L17/L18/L8_TIMING/L9 batch synth (R46 relocated)
[14c4/15]	L1/L2/L6/L7/L12 universal protocol doc facts (R50)
[14c5/15]	L4/L5/L10/L14/L15 residual cleanup (R52)
[14d/15]	L19-L23 skeleton emit
[14e/15]	serial_peripheral_protocol class synth (R53/R54/R55) — 81-protocol detector→synth dispatch
[14e2/15]	bus_interconnect_protocol Tier-2 synth (TileLink/Wishbone/Avalon/OCP/AXI-Stream)
[14e3/15]	Universal packet/PDU L10↔L3 opcode-consistency sweep
[15/15]	Coverage / parity report

→ **34 distinct Phase-1 markers** (15 main + 19 sub/overlay/dispatch).

A.2 Phase 2 — step functions (phase2_one_shot_runner.py) — global steps 35-49

Continuous global numbering (does NOT restart at 1; continues from Phase 1's 34 markers). Auto-generated live in FLOW_STEPS_GENERATED.md.

#	Step	Note
35	step_phase1	re-run/ingest Phase 1 if needed
36	step_rig_topology_skeleton	topology scaffold
37	step_rtl_gen	WAIVE for rtl_gen=null ic_class → spec-to-rtl role; phase2_scaffold_gen.py emits top/regs/fsm/tb/soc_wrap/cocotb
38	step_full_stack_tb_gen	self-checking TB
39	step_reference_tb	reference-TB conformance (eco_loop ≤ 3 retries)
40	step_yosys_synth	gate-level synth
41	step_qsf_gen	FPGA project
42	step_sdc_gen	constraints
43	step_otp_image_check	OTP image (if any)
44	step_fpga_compile	FPGA build → .sof
45	step_fpga_burn	program board
46	step_usb_hid_tester_verify	host protocol-tester acceptance
47	step_emit_phase2_manifests	manifests
48	step_final_audit	aggregate audit
49	step_phase3	chains into Phase 3 when run end-to-end

Surrounding gates: rtl_hygiene_lint --fix, spec_conformance_check, chip_top_gate_wrapper_gen, MCP_eda_lint/eda_synth/eda_cocotb.

A.3 Phase 3 — step functions (`phase3_one_shot_runner.py`) — global steps 50-55 (NOT 1-6)

Phase 3 is the back-end; its step functions continue the global count — they do **not** restart at 1.

#	Step	Tool (open-source)
50	<code>step_synth</code>	yosys (+ tie-cell pass: <code>setundef -zero</code> ; <code>hilo</code> map; <code>splitnets</code> ; <code>clean</code>)
51	<code>step_pnr</code>	OpenROAD (floorplan→PDN→place→CTS→route)
52	<code>step_gds</code>	KLayout (<code>def2gds</code>)
53	<code>step_drc</code>	KLayout sky130 deck
54	<code>step_lvs</code>	netgen / yosys_equiv — the LVS sign-off chain (§ C)
55	<code>step_canonicalize_artefacts</code>	normalize outputs (+ the § D sign-off emitters)

A.4 Analog A1–A8 (`analog_one_shot_runner.py` , parallel to Phase 2)

Step	Name	Output
A1	<code>spec_extract</code>	<code>analog/<block>/A1_spec.json</code>
A2	<code>topology_select</code>	<code>A2_topology.json</code>
A3	<code>netlist_gen</code>	<code><block>.sp</code>
A4	<code>corner_sweep</code>	<code>A4_corners.json</code>
A5	<code>layout</code> (Magic)	<code>A5_layout.json</code> (needs DRC-clean + LVS-match flags)
A6	<code>post_layout_resim</code>	<code>A6_postsim.json</code>
A7	<code>hardmacro_gen</code>	<code>{.lef, .lib, .gds, .v}</code> → feeds Phase 3
A8	<code>hw_verify</code> (HIL)	<code>A8_hw_verify.json</code>

A.5 Mixed-signal M1–M4 (skill-level, no dedicated runner)

Step	Name	Where
M1	<code>top merge</code>	<code>mixed_signal_m1_top_merge_check.py</code> + <code>skill mixed-signal-cosim</code>
M2	<code>co-sim setup</code>	<code>skill mixed-signal-cosim</code>
M3	<code>co-sim run</code>	<code>skill mixed-signal-cosim</code>
M4	<code>integration verify</code>	<code>skill mixed-signal-cosim</code>

B. 33-STEP STAGE MODEL — ALL 33 STEPS, CONTIGUOUS (sign-off view, `33_step_flow_overview.md`)

Every step 1→33, no gaps. Stage is a column (S1 RTL+verify / S2 synth+DFT / S3 physical+sign-off / S4 output+validation). (§ A is the runner-marker view; § D below is a subset — only the sign-off checks with a known open-source gap.)

#	Stage	Step	Tool/Skill	Gate
1	S1	Spec-to-RTL	<code>spec-to-rtl</code>	L1-L9 present + RTL emitted

#	Stage	Step	Tool/Skill	Gate
2	S1	Lint	eda_lint + Phase-2a gates + polluter check	Verilator 0 errors
3	S1	CDC / RDC check	cdc-check	CDC paths synchronised
4	S1	Simulation	testbench-gen + eda_simulate + coverage	all tb PASS + coverage
5	S1	Formal verification	formal-verify + assertion-gen	k-induction proved (informational waiver if no model)
6	S1	FPGA early prototype	fpga-test-harness + eda_fpga_compile/program + on-board BIST	.sof + BIST PASS
7	S2	Constraint setup	constraint-gen	*.sdc + 3-corner pvt_matrix.json
8	S2	SDC validation	SDC lint	all clock/IO constraints present
9	S2	Synthesis (Yosys)	eda_synth + synth-doctor (+ tie-cell pass)	mapped netlist + cell count
10	S2	Pre-layout STA	eda_sta_mcorner (SS/TT/FF)	WNS/WHs all corners
11	S2	DFT insertion (scan + ATPG)	dft-insert + atpg + eda_dft	scan chain + stuck-at ≥85%
12	S2	Post-DFT optimization	resynth / buffering	timing held
13	S2	Equivalence check	equivalence-check + Yosys equiv	RTL ≡ post-DFT netlist
14	S3	Floorplan + PDN	eda_pnr (init)	area / utilization
15	S3	Clock planning	clock-planning skill	clock-buffer list
16	S3	Placement (global + detailed)	eda_pnr	placement legal
17	S3	CTS	eda_pnr enable_cts=true	clock skew
18	S3	Post-CTS hold fixing	repair_timing -hold	WHS > 0
19	S3	Routing (global + detailed)	eda_pnr enable_detailed_route=true + def_stage_progression_check	0 overflow + DEF SHA differs
20	S3	Parasitic Extraction (RC→SPEF)	eda_extraction (OpenRCX)	spef_extraction_check (sky130 = no captable → ENV-BLOCKED)
21	S3	Post-route STA (MMMC)	eda_sta_mcorner	3-corner pass
22	S3	IR Drop	OpenROAD PSM analyze_power_grid	ir_drop_report_check (FIXED v0.2.4)
23	S3	EM check	PSM -enable_em	em_report_check (FIXED v0.2.4)
24	S3	Antenna check	OpenROAD check_antennas	0 violation (report-path FIXED v0.2.4)
25	S3	Signal Integrity (crosstalk/ noise)	SI screen (decoupled-C)	si_crosstalk_check (screen; full needs SPEF)
26	S3	Post-Layout Gate-Level Sim (+SDF)	eda_simulate	post_layout_sim_check PASS

#	Stage	Step	Tool/Skill	Gate
27	S3	Physical Verification	eda_drc_klayout + eda_lvs + ERC	DRC=0 / LVS device-exact / ERC floating-net
28	S3	ECO repair loop	eco-plan	eco_loop_audit PASS
29	S4	Power analysis	pre + post layout	meets spec
30	S4	Metal Fill (density fill)	OpenROAD filler_placement → filled.def	metal_fill_density_check (FIXED v0.2.4)
31	S4	Tapeout checklist	tapeout-checklist + signoff_audit	4/4 strict
32	S4	GDSII output	eda_gds + def2gds	only if step 27 clean
33	S4	FPGA final sign-off	recompile + on-board test + fpga_on_board_attestation_check	bitstream hash + hw evidence

Numbering caveat: the Stage-3 sign-off checks ALSO carry a sign-off-audit numbering used in § D (SPEF 22 / STA 23 / IR 24 / EM 25 / Antenna 26 / SI 27 / DRC-LVS-ERC 30 / fill 33), which differs from THIS 33-step numbering above (SPEF 20 / STA 21 / IR 22 / EM 23 / Antenna 24 / SI 25 / PV 27 / fill 30). Same checks, two ID schemes; to be unified by flow_doc_emit.py.

C. The LVS sign-off chain (under Phase-3 step_lvs, NEW v0.1.96→v0.2.2)

Layer	What	Tool
1	Structural LEC (default)	eda_lvs mode=yosys_equiv (equiv_simple + equiv_induct) — SAT-model unproven = Category-D gap
2	Device-level coverage	eda_extraction (magic ext2spice) + eda_lvs mode=netgen + lvs_netgen_setup_emit.py
3	Powered-netlist closure	OpenROAD write_verilog - include_pwr_gnd (after global_connect)
4	Top-level port labels	Route A magic_port_extract_emit.py (port makeall) / Route B lvs_def_port_seed.py (DEF seed)
5	Sign-off guard (MANDATORY)	lvs_signoff_guard.py — RAISES on a portless / vacuous match

D. Phase-3 sign-off checks — gap status (SUBSET; sign-off-audit numbering)

This is a SUBSET, not the full step list — only the Phase-3 sign-off checks that had an open-source gap (which is why 14-21 / 28 / 29 / 31 / 32 are absent here: they had no gap). The complete contiguous 1→33 list is § B. Status as of v0.2.4 (fixes shipped — see backlog ORGANIC-20260531-phase3-signoff-chain-open-source-gaps):

Step	Check	Status (v0.2.4)	Severity
22	SPEF (OpenRCX)	WORKS (prior "ENV-BLOCKED" was a false negative) — sky130A does ship the OpenRCX captable at /foss/pdks/sky130A/libs.tech/openlane/rules.openrcx.sky130A.	● works

Step	Check	Status (v0.2.4)	Severity
		<code>{min,nom,max}.magic;</code> <code>extract_parasitics -</code> <code>ext_model_file <...nom.magic></code> + <code>write_spef</code> extracts on a real routed DEF (spm: 1370 rc segments, 330 nets, 1700 caps). The earlier RCX-0107 "0 segments" was an EMPTY (routing-less) DEF, not a missing captable.	
23	Post-route STA (MMMC)	passes; pilots report slack +X ns MET	● none
24	IR drop (PSM)	cascading-missing on SPEF	◆ medium
24	IR drop	FIXED — OpenROAD PSM <code>analyze_power_grid</code> (walks DEF SPECIALNETS directly; no SPEF needed — the cascade premise was wrong) → <code>reports/phase3/ir_drop.{rpt,json}</code>	● fixed
25	EM	FIXED — PSM <code>-enable_em</code> → <code>em.{rpt,json}</code>	● fixed
26	Antenna	FIXED — <code>check_antennas</code> re-emitted to <code>antenna.{rpt,json}</code> (report-path)	● fixed
27	SI (crosstalk)	PARTIAL — decoupled-C SCREEN_PASS now emitted; full coupling-cap SI still needs SPEF (env-blocked)	◆ screen
30	DRC / LVS / ERC	PARTIAL — KLayout sky130 DRC + Magic floating-net ERC + device-level LVS (§ C) all wired/ passing; full Calibre PERC (latch-up/ESD) env-deferred	◆ partial
33	Metal fill	FIXED — OpenROAD <code>filler_placement</code> → <code>phase3/stage3/pnr/filled.def</code> + <code>density.{rpt,json}</code>	● fixed
18	Spare cells	FIXED — <code>spare_cells.json</code> now has <code>rows[]</code> (derived from existing placement; placement unchanged)	● fixed
5	Formal	confirmed INFORMATIONAL waiver (altsyncram no model) — no code change	● none

None is a circuit-design error — all script-ordering / cascading / environment / report-schema. Actionable fixes tracked in `ORGANIC-20260531-phase3-signoff-chain-open-source-gaps`.

E. Totals

View	Count
Phase 1 runner markers	34 (15 main + 19 sub)
Phase 2 step functions	14 (+1 phase3 hook)
Phase 3 step functions	6

View	Count
Analog	A1–A8 (8)
Mixed-signal	M1–M4 (4)
33-step Stage model	33 (Stage 1:6 / 2:7 / 3:15 / 4:5)
LVS sign-off chain layers	5
Phase-3 sign-off checks	10

F. Keep current

This is a **derived** enumeration. The § A runner-marker tables are now auto-generated by `programs/flow_doc_emit.py` (shipped v0.2.3) into `FLOW_STEPS_GENERATED.md` — run `python3 flow_doc_emit.py` after any runner change; `flow_doc_emit.py --check` (`tests/test_flow_doc_emit.py`) fails CI on drift. The § B 33-step Stage model, § C LVS chain, and § D sign-off table remain hand-curated (not derivable from markers); unifying the two Stage-3 numberings into the generator is the remaining follow-up.